Department of CSE

Mini Project Report

" Malware Family Classification Using a CNN with Attention on the MaleBin Dataset "

Course Code: ICE 478/CSE475

Section: 2

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1. Introduction

Malware is one of the major cybersecurity threats in today's digital world. It can damage computer systems, steal sensitive information, and disrupt normal operations. As new malware variants are continuously being developed, traditional signature-based detection methods often fail to identify previously unseen malware. Therefore, researchers are increasingly using machine learning and deep learning techniques for automatic malware detection and classification.

One popular approach is to convert malware binary files into grayscale images and apply image classification techniques. Malware samples from the same family usually produce similar visual patterns, allowing deep learning models such as Convolutional Neural Networks (CNNs) to learn meaningful features directly from the images. In this project, the MaleBin Malware Binary Greyscale Images dataset is used to classify different malware families. Since the project follows Track 3, a CNN with an attention mechanism will be developed in the later stages. Before building the model, it is important to understand the dataset through Exploratory Data Analysis (EDA). This report presents the EDA conducted on the MaleBin dataset and summarizes the key observations obtained from the analysis.

2. Dataset Overview

The MaleBin dataset contains malware binaries that have been converted into grayscale images. Each malware image belongs to a specific malware family, and the images are organized into separate folders according to their labels. The dataset is designed for malware family classification using computer vision techniques.

Unlike conventional malware datasets that require manual feature extraction, this dataset enables deep learning models to automatically learn features from image patterns. However, because malware binaries have different sizes, the generated images also vary in their dimensions. Therefore, preprocessing steps such as image resizing and normalization are necessary before model training.

Dataset Information

- **Dataset Name:** MaleBin Malware Binary Greyscale Images
- **Data Type:** Grayscale Images
- **Image Format:** PNG
- **Classification Type:** Multi-class Malware Family Classification
- **Source:** Kaggle

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the overall characteristics of the dataset before training a deep learning model. The analysis focused on image visualization, class distribution, image dimensions, and pixel intensity patterns. These analyses help identify potential issues such as class imbalance, inconsistent image sizes, and preprocessing requirements.

3.1 Dataset Description

The MaleBin dataset is organized into separate folders, where each folder represents a different malware family. Each folder contains multiple grayscale images generated from malware binary files, and each image belongs to the malware family indicated by its folder name. This folder-based organization makes it convenient to assign class labels during data loading and model training.

Before starting the analysis, the dataset structure was verified to ensure that all images were correctly placed in their respective folders and could be accessed without errors. Understanding this organization is important because it forms the basis for preprocessing, data loading, and supervised learning.

	Property	Value
0	Dataset name	MaleBin (Malware Binary Greyscale Images)
1	Source	Kaggle (tashiee/malebin-malware-binary-greysca...
2	Application area	Malware family classification from binary-to-i...
3	Total images	12464
4	Number of classes	39
5	Image format	PNG/JPG greyscale
6	Balanced?	TBD — see class balance below

Figure 3.1: Dataset Folder Structure

3.2 Sample Malware Images

Random malware images from different classes were visualized to observe their appearance. Although the images are generated from binary files, they exhibit unique texture patterns that differ across malware families. Some images contain dense repetitive structures, while others display irregular patterns.

Visual inspection helps confirm that the images are loaded correctly and provides an initial understanding of the visual differences among malware families.

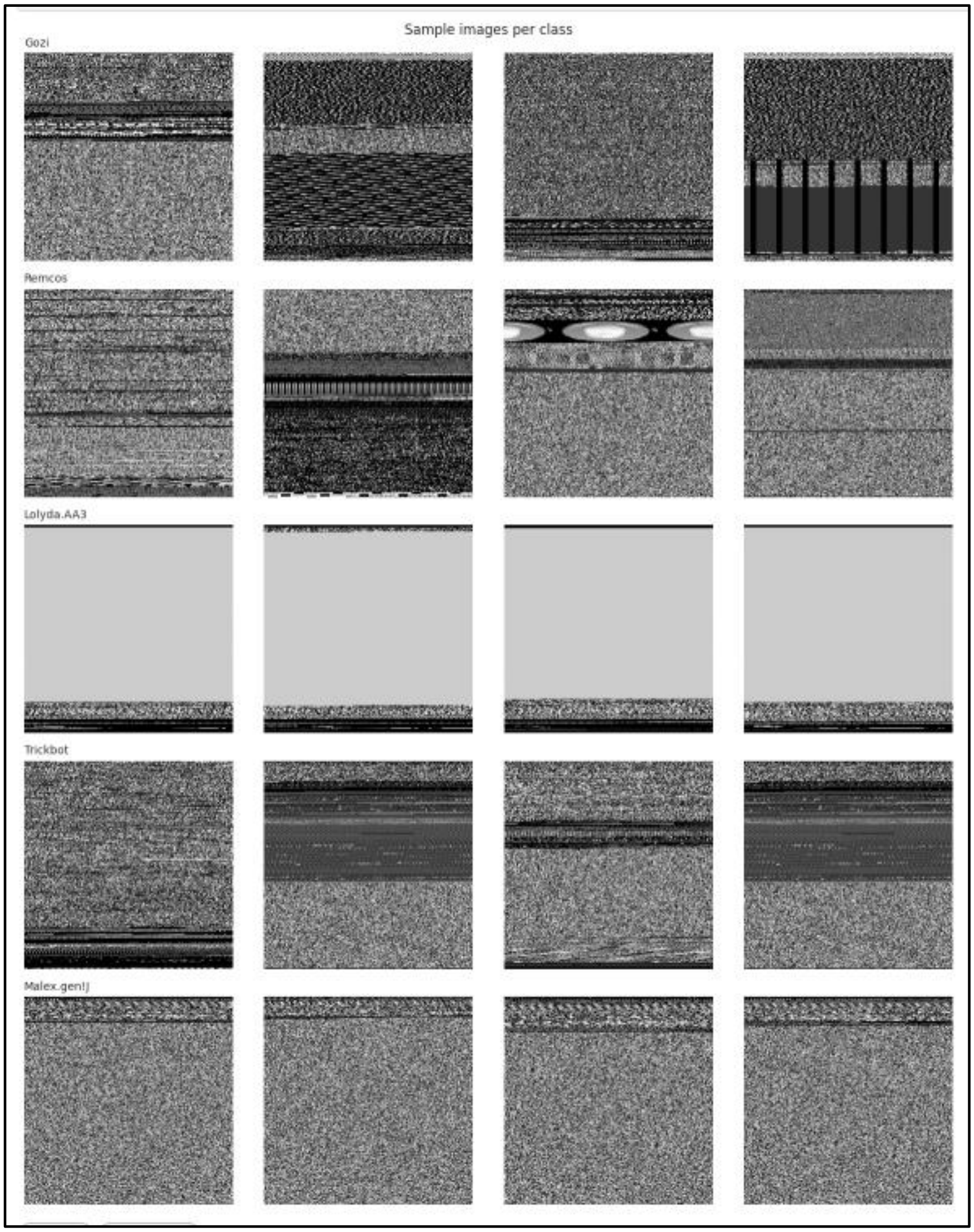

Figure 3.2: Sample Malware Images from Different Malware Families

3.3 Class Distribution

The number of samples in each malware family was analyzed to understand the distribution of the dataset. A bar chart was created to compare the number of images available for each class.

The analysis shows whether the dataset is balanced or imbalanced. A balanced dataset allows the model to learn equally from all classes, while an imbalanced dataset may bias the model toward classes with more samples. If necessary, techniques such as data augmentation or class weighting can be applied during model training.

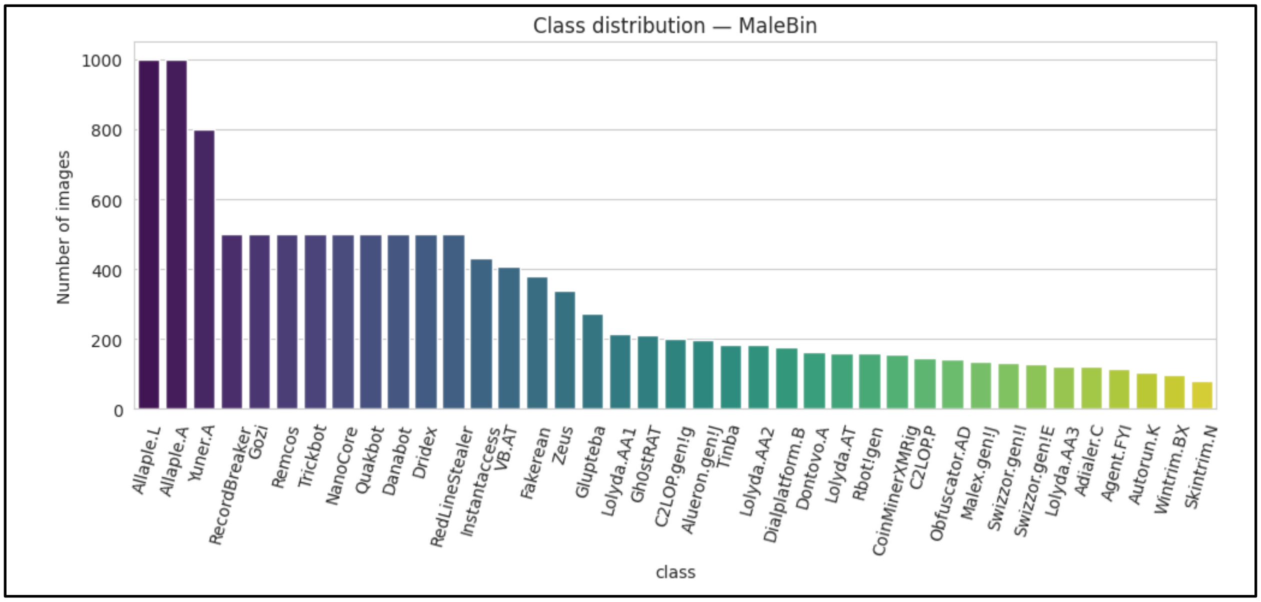

Figure 3.3: Class Distribution of Malware Families

3.4 Image Dimension Analysis

The dimensions of all malware images were analyzed to determine whether they have consistent sizes. Since malware binaries vary in size, the generated images also have different widths and heights.

The analysis indicates that image resizing will be required before training the CNN model. Using a fixed image size ensures that all images can be processed efficiently by the deep learning model.

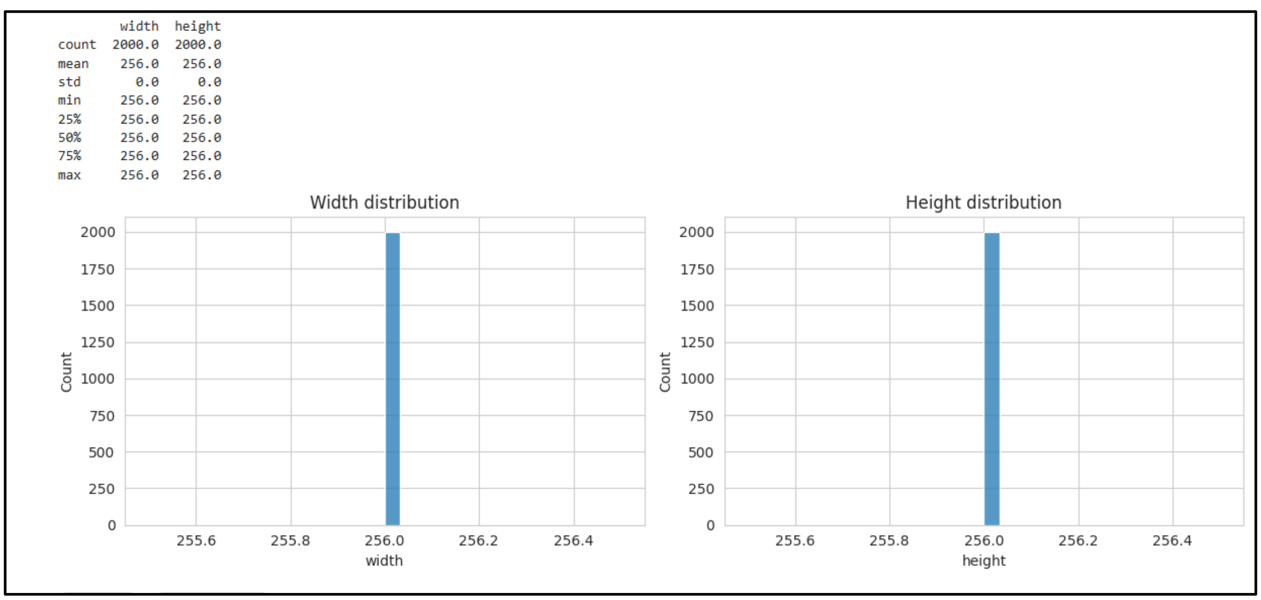

Figure 3.4: Distribution of Image Dimensions

3.5 Pixel Intensity Distribution

Pixel intensity analysis was performed to examine the grayscale value distribution of the malware images. Since the images are represented in grayscale, each pixel has an intensity value ranging from 0 (black) to 255 (white). A histogram was generated to visualize the frequency of these intensity values across the dataset.

The histogram shows that most pixel values are concentrated in the lower and middle intensity ranges, indicating that the malware images contain more dark regions than bright regions. This information is useful for understanding the visual characteristics of the dataset and deciding whether normalization is required before model training.

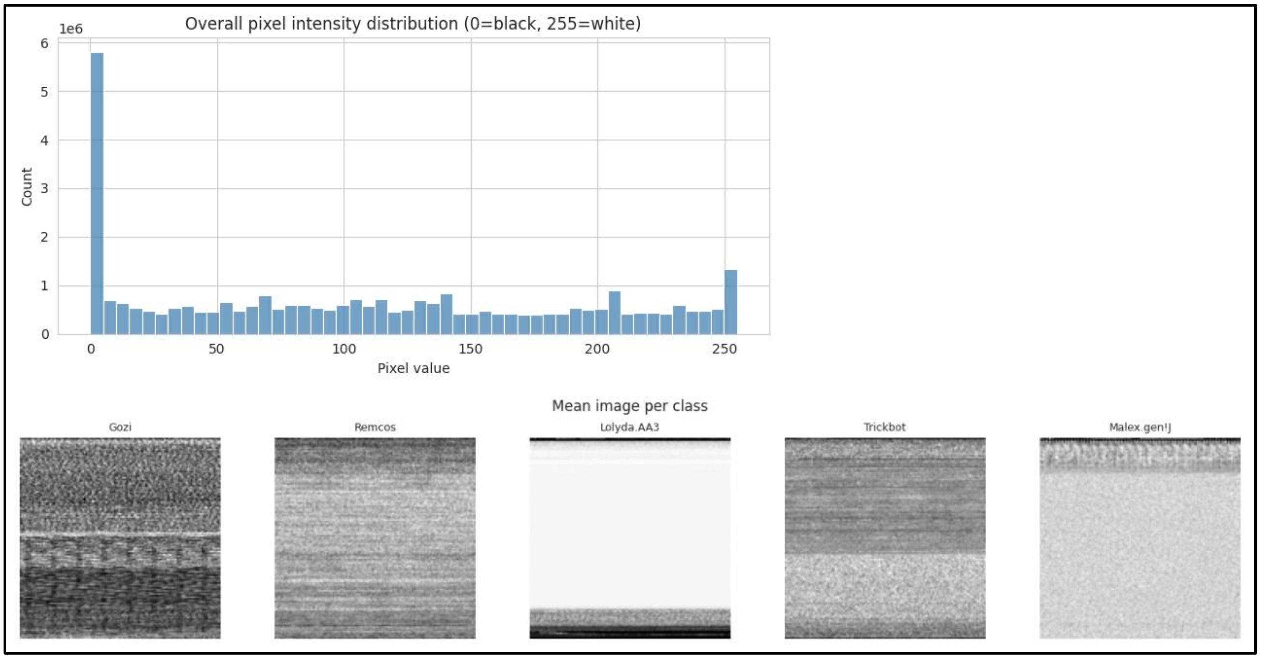

Figure 3.5: Pixel Intensity Distribution of the MaleBin Dataset

3.6 Image Metadata Analysis

After loading the dataset, metadata was extracted from each malware image to better understand its basic properties. The analysis included image mode, width, height, file size, and aspect ratio. This information helps verify the consistency of the dataset before model development.

The results showed that the malware images have consistent image modes and similar structural properties. Examining image metadata is an important preprocessing step because it helps identify unexpected image formats or inconsistencies that could affect model training.

label	width	height	aspect_ratio	mode	pixels	filename	file_size_kb	label_id	cnn_ready	suggested_size	nr
Malware_Merge/Adialer.C	256	256	1.0	L	65536	0528185d9960c8bd1b6692d2ccc8b9dd.png	43.902344	0	True	224x224	
Malware_Merge/Adialer.C	256	256	1.0	L	65536	050f962b76f3b115044873d9e3a07a97.png	43.903320	0	True	224x224	
Malware_Merge/Adialer.C	256	256	1.0	L	65536	053f41978f89e436244c691072f4a548.png	43.495117	0	True	224x224	
Malware_Merge/Adialer.C	256	256	1.0	L	65536	06e21af7e2fdd67622d0122e113b0d53.png	43.627930	0	True	224x224	

Figure 3.6: Summary of Image Metadata

3.7 Pixel Feature Distribution

To better understand the visual characteristics of the malware images, several statistical pixel features were extracted and analyzed. These include the mean pixel intensity, standard deviation, entropy, and the proportion of bright pixels. Histograms were generated to visualize the distribution of these features across the dataset.

The feature distributions provide useful information about the texture and intensity variations among malware samples. These statistics help understand the overall characteristics of the dataset and provide useful insights before developing the CNN model.

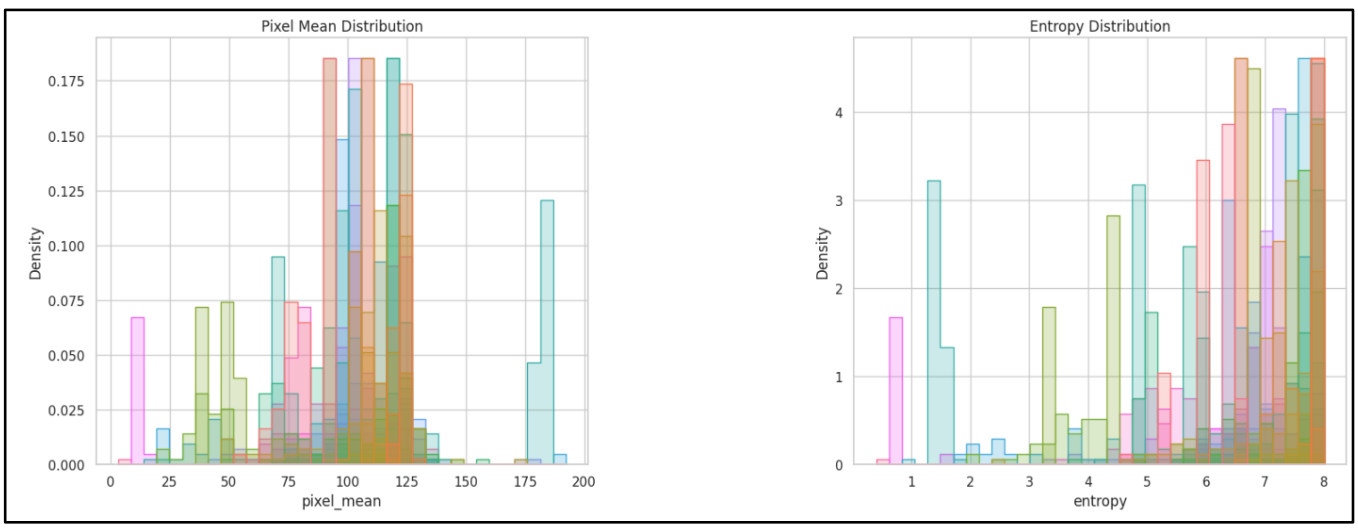

Figure 3.7: Distribution of Extracted Pixel Features

3.8 Correlation Heatmap of Pixel Features

A correlation analysis was performed to examine the relationships among the extracted pixel features. The correlation matrix was visualized using a heatmap, where different colors indicate the strength of positive or negative relationships between features.

The heatmap helps identify highly correlated features that may contain similar information. Understanding these relationships provides a better overview of the dataset and supports future feature analysis during model development.

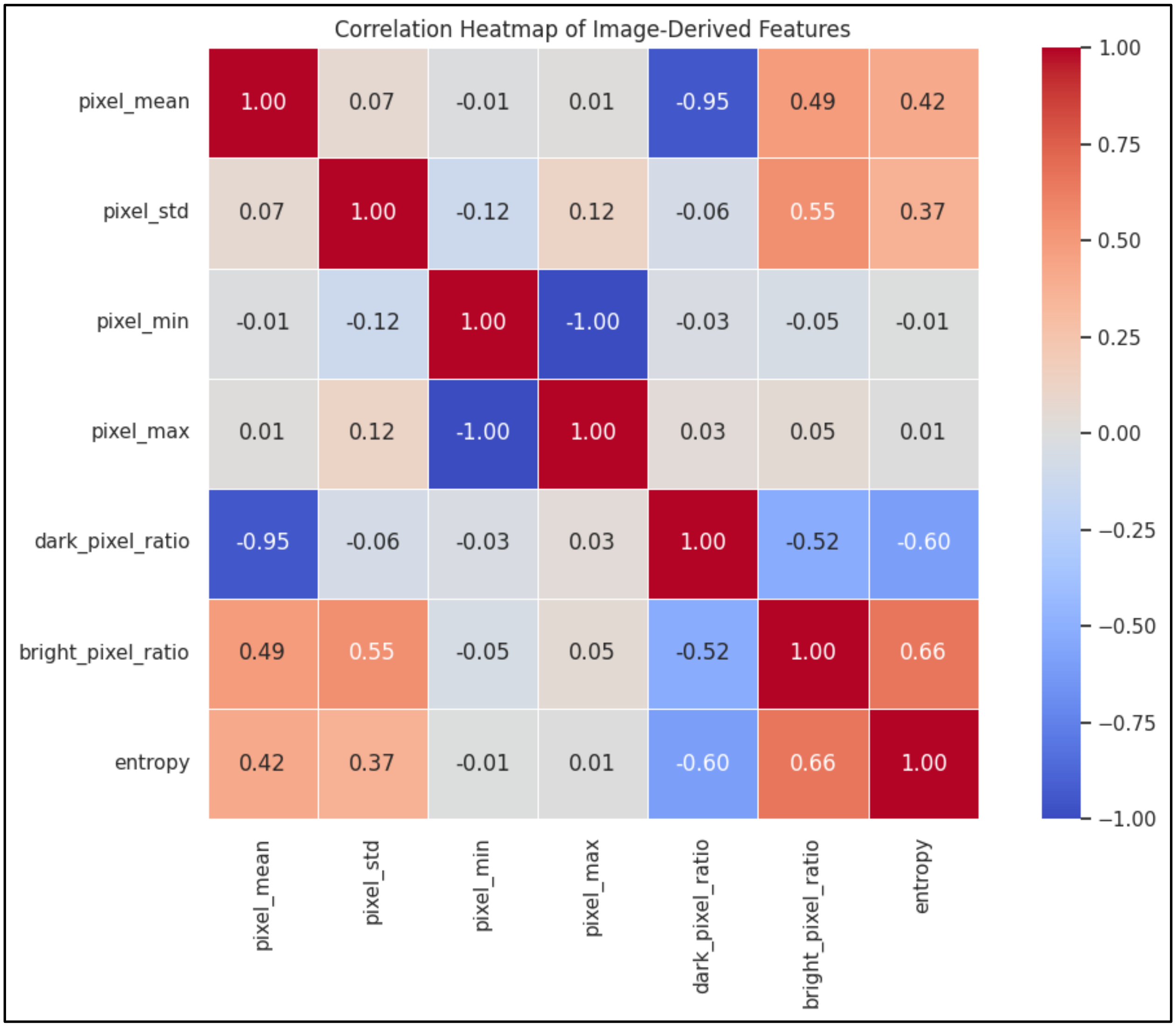

Figure 3.8: Correlation Heatmap of Pixel Features

3.9 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the extracted pixel features while preserving the most significant information. The first two principal components were visualized to observe the distribution of malware samples in a lower-dimensional feature space.

The PCA visualization provides an overview of how different malware families are distributed based on their extracted image features. Although some overlap exists, several malware families form distinguishable clusters, indicating that the extracted features contain useful discriminatory information.

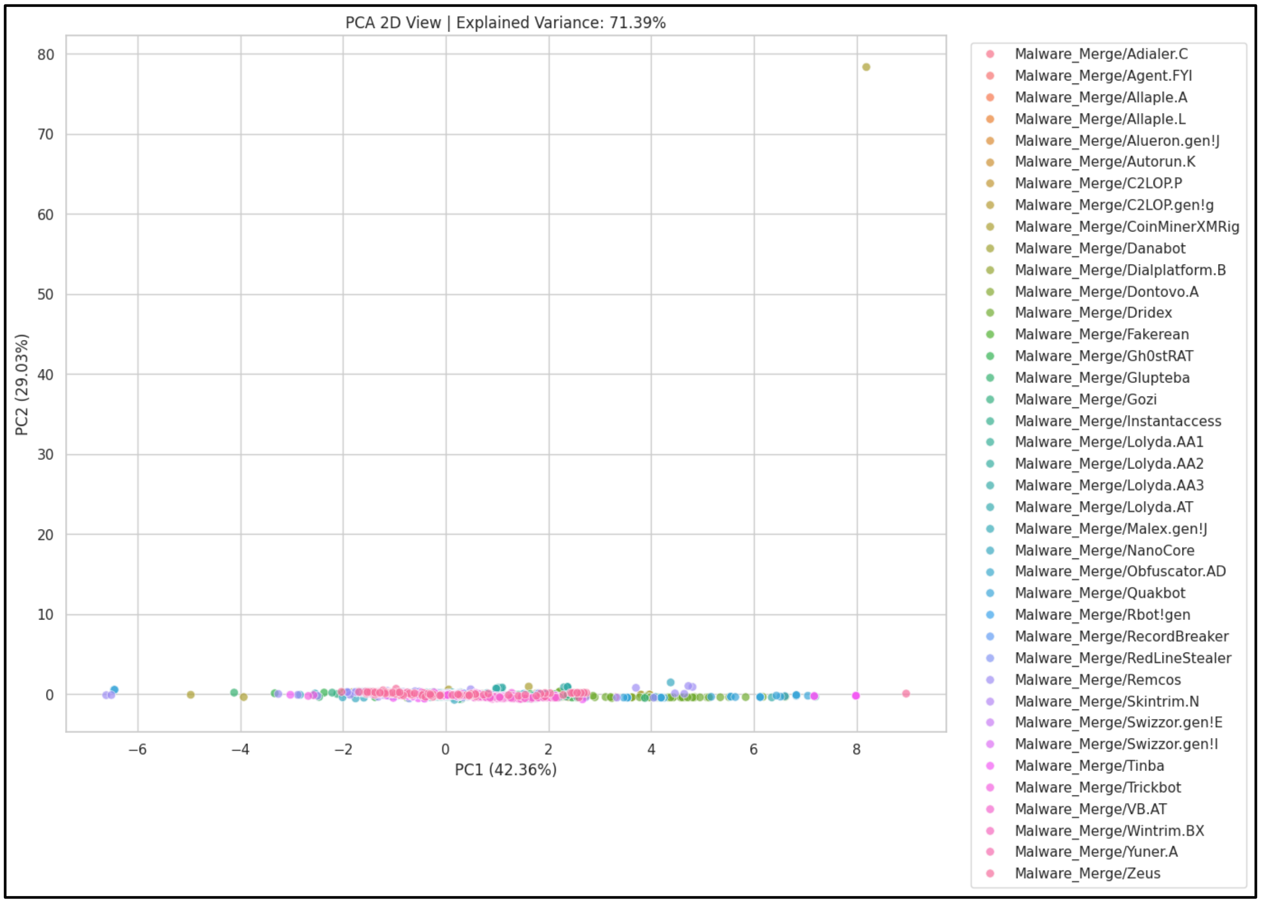

Figure 3.9: PCA Projection of Malware Samples

Key Observations

The exploratory data analysis revealed several important characteristics of the MaleBin dataset.

- The dataset is well organized into different malware families.
- Malware images from different families show distinct visual patterns.
- The dataset has a moderate class imbalance.

- Image dimensions vary, making resizing necessary before training.
- Pixel intensity analysis shows that most images contain darker regions.
- Metadata and feature analysis indicate that the dataset is consistent and suitable for deep learning.
- The correlation heatmap and PCA visualization show that the extracted features contain useful information for distinguishing malware families.

4. Conclusion

This report presented an exploratory data analysis of the MaleBin dataset. The analysis included dataset organization, image visualization, class distribution, image dimension analysis, pixel intensity analysis, metadata analysis, feature extraction, correlation analysis, and PCA.

The results show that the dataset is suitable for image-based malware classification. Although some preprocessing, such as image resizing, will be required, the extracted features provide meaningful information for distinguishing malware families. Overall, the EDA provides a solid foundation for developing the proposed CNN with an attention mechanism in the next phase of the project.